1 Hyper-Detailed Pedagogical Blueprint Real Analysis: Change of Variables & Rigorous Foundations

Director's Cut & AI Extraction Protocol
Lecture Segment: 00:00 – 15:00

Insight: The "Wall" of Multivariable Calculus

This segment serves as the bridge between computational calculus and rigorous Real Analysis. The professor is actively shifting the students' mindset away from "memorizing integration formulas" and towards "understanding the geometric distortions of space via linear algebra (Jacobians)." Notice how carefully he handles domains and inverse mappings—this rigor is the core of the discipline.

2 Reminder: The Setup for the Change of Variables

Cleaned Spoken Protocol [Time: 00:00:11 - 00:01:29]

I hope you all had a great holiday and have recovered from the first half of the semester so we can continue with our work! Let me start by reminding you of what we did last time. In our previous lecture, we stated and began to prove what we playfully call the "final boss of integration": the Change of Variables formula. Let us review the setup.
As always, we begin with our domain $U \subset \mathbb{R}^n$. We apply a diffeomorphism—which we call Φ—that maps U to another domain V.

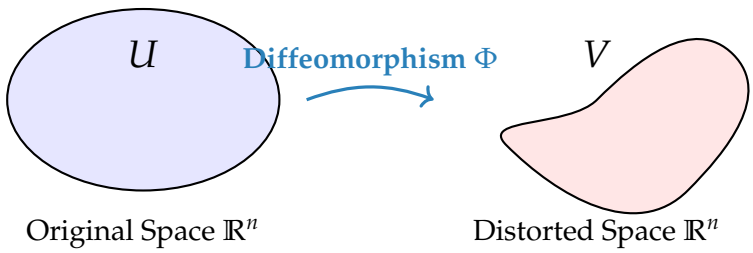
Board State & Physical Actions: Board State 1: The Diffeomorphism

The professor writes the title "Change of vars." and begins defining the mathematical environment on the far left board. He writes the mapping $U \to V$ and explicitly writes "diffeo" in orange chalk beneath the arrow.

Mathematical Setup (Stroke-by-Stroke): Let $U, V \subset \mathbb{R}^n$. We define the mapping between these spaces:

$$U \xrightarrow[\text{diffeo}]{\Phi} V$$

Redundant Explanation of Step: We establish a transformation Φ mapping a domain U to a codomain V. The orange annotation "diffeo" indicates this mapping is a *diffeomorphism*. A diffeomorphism is a heavily constrained, "well-behaved" mapping: it must be a bijection (one-to-one and onto), it must be differentiable everywhere, and its inverse Φ^{-1} must also be differentiable everywhere. This ensures the space is smoothly warped without tearing or folding.



Cleaned Spoken Protocol [Time: 00:01:29 - 00:02:45]

We then consider a Jordan measurable set $A \subset \mathbb{R}^n$, with the strict condition that its closure, $\overline{A}$, must be entirely contained within the domain U of our diffeomorphism. Next, we need a continuous function to integrate. We define a function f mapping from the closure $\overline{A}$ to the real numbers $\mathbb{R}$.

Board State & Physical Actions: Board State 2: Domain Correction

The professor writes $A \subset \mathbb{R}^n$ J. m. st. $\overline{A} \subset U$. When defining the function f, he initially says "from U or from A to $\mathbb{R}$", but immediately catches his imprecision and writes $\overline{A} \to \mathbb{R}$ continuous.

Mathematical Setup (Continued): We restrict our attention to a specific subset of U:

$$A \subset \mathbb{R}^n \text{ is Jordan measurable, such that } \overline{A} \subset U$$

We define our integrand function:

$$f : \overline{A} \to \mathbb{R} \quad \text{(continuous)}$$

Redundant Explanation of Step: Why the closure $\overline{A}$? By requiring the *closure* of A to be strictly contained within the open set U, we ensure that the boundary of A behaves nicely under the mapping Φ, and we avoid pathological singularities that could exist on the absolute edge of the domain U. We then ensure f is continuous over this closed, bounded region, guaranteeing Riemann integrability.

3 The Clunky Formulation of the Theorem

Cleaned Spoken Protocol [Time: 00:02:45 - 00:03:50]

With this setup, the theorem states that the integral of $f(x)$ over the region A is equivalent to the integral evaluated over the mapped image, $\Phi(A)$. However, to calculate this, we must evaluate our function f at the pre-image, $\Phi^{-1}(y)$. Furthermore, the differential element dy must be corrected by the distortion of the volume, which is exactly the absolute value of the determinant of the Jacobian of Φ, evaluated at $\Phi^{-1}(y)$.

Board State & Physical Actions: Board State 3: The Clunky Formula

The professor writes out the full Change of Variables formula exactly as it was derived in the previous lecture. It is notationally heavy due to the inverse mappings inside the arguments.

The Initial Theorem (Inline Annotated):

$$\underbrace{\int_A f(x)\,dx}_{\text{Integral in Original Space}} = \int_{\Phi(A)} f(\underbrace{\Phi^{-1}(y)}_{\text{Pre-image under } \Phi}) \underbrace{\frac{1}{|\det J\Phi(\Phi^{-1}(y))|}}_{\text{Correction for Volume Distortion}} dy \tag{1}$$

Redundant Explanation of Step: To evaluate the integral in the distorted space V (where the variable is y), every x in the original function must be replaced with the mapping that takes y back to x, which is $\Phi^{-1}(y)$. Because the transformation Φ stretches and squishes space, the resulting "volume pixels" (dy) are distorted. To preserve the total "mass" of the integral, we must divide by the scaling factor of that distortion. The scaling factor of a linear transformation

is the determinant of its matrix—hence, we divide by the determinant of the Jacobian matrix of Φ.

> ***Cleaned Spoken Protocol [Time: 00:03:50 - 00:06:50]***
>
> *I previously explained the geometric intuition behind why this theorem is true, but we didn't have time to complete the formal proof. I propose that we leave the proof for now. Instead, we will look at some concrete examples to see how we apply this theorem in actual calculations. Once you see it in action, you will understand why it is worth the effort to prove it. We will revisit the formal proof in two weeks during our revision buffer week.*
> **(The professor then spends a few minutes discussing an upcoming interactive teaching session on Thursday with guest coaches from ETH. He asks students to attend in person for the interactive exercises.)**

4 Rewriting the Theorem via the Inverse Map

> ***Cleaned Spoken Protocol [Time: 00:06:50 - 00:08:20]***
>
> *Let us rewrite this theorem into a format that is much easier to use in practice. Instead of constantly writing Φ^{-1} and dragging it through our equations, let us define a new mapping Ψ to be exactly equal to the inverse, Φ^{-1}.*
> *If we substitute this into our theorem, the integration region $\Phi(A)$ remains, but the function evaluation $f(\Phi^{-1}(y))$ simply becomes $f(\Psi(y))$. But what happens to that clunky determinant in the denominator?*

Board State & Physical Actions: Board State 4: Defining Psi

The professor moves to the center board to clean up the notation. He defines Ψ and prepares to algebraically manipulate the Jacobian fraction.

Simplifying the Notation: Let us define the inverse map explicitly:

$$\Psi = \Phi^{-1}$$

Substituting this directly into Equation 1, we get an intermediate step:

$$\int_A f(x)\,dx = \int_{\Phi(A)} f(\Psi(y)) \frac{1}{|\det J\Phi(\Psi(y))|}\,dy$$

> ***Cleaned Spoken Protocol [Time: 00:08:20 - 00:10:45]***
>
> *Remember the lemma we proved regarding the differential of an inverse function. If we compose Φ with its inverse Ψ, we get the identity map.*
> *By applying the chain rule, the differential of this composition evaluated at a point y is the differential of Φ multiplied by the differential of Ψ. Because the differential of the identity map is just the identity matrix, this means that the Jacobian matrix of Φ multiplied by the Jacobian matrix of Ψ equals the unit (identity) matrix.*
> *Because the determinant is a multiplicative property for matrices, the determinant of one matrix must be the exact inverse (reciprocal) of the other. Therefore, instead of dividing by the determinant of $J\Phi$, we can simply multiply by the determinant of $J\Psi$!*

Board State & Physical Actions: Board State 5: Chain Rule Derivation

The professor formally derives the relationship between the determinants on the board, line by line, mapping the abstract function composition to concrete matrix algebra.

Derivation of the Jacobian Reciprocal Relation:

$$\underbrace{\Phi \circ \Psi = \mathrm{id}}_{\text{Function Composition}} \implies \underbrace{D\Phi_{\Psi(y)} \circ D\Psi_y = \mathrm{id}}_{\text{Taking the Differential (Chain Rule)}}$$
$$\implies \underbrace{J\Phi(\Psi(y))}_{\text{Matrix 1}} \cdot \underbrace{J\Psi(y)}_{\text{Matrix 2}} = \underbrace{I_n}_{\text{Identity Matrix}}$$

Redundant Explanation of Step: The chain rule in multivariable calculus dictates that the derivative of a composition of functions is the matrix product of their Jacobian matrices. Since the composition yields the identity function ($y \mapsto y$), the product of their Jacobians must yield the identity matrix I_n.

Taking the determinant of both sides, and using the property that $\det(AB) = \det(A)\det(B)$ and $\det(I_n) = 1$:

$$\det\Big(J\Phi(\Psi(y)) \cdot J\Psi(y)\Big) = \det(I_n)$$
$$\det(J\Phi(\Psi(y))) \cdot \det(J\Psi(y)) = 1$$
$$\implies \underbrace{\frac{1}{\det J\Phi(\Psi(y))}}_{\text{Clunky Denominator}} = \underbrace{\det J\Psi(y)}_{\text{Clean Numerator}}$$

Insight: The Power of Linear Algebra

The professor takes a moment here to show how beautifully linear algebra serves calculus. The geometric reality that "stretching space forward by factor k, then pulling it backward, returns it to normal" is flawlessly encoded in the algebraic fact that the determinants of inverse matrices multiply to 1.

Cleaned Spoken Protocol [Time: 00:10:45 - 00:12:15]

This gives us a highly practical, informal way to remember the substitution rule. Just like in Analysis 1, when you substitute $x = \Psi(y)$, you must replace the differential. But instead of replacing dx with the 1D derivative $\Psi'(y)\,dy$, you replace it with the absolute determinant of the Jacobian matrix, $|\det J\Psi(y)|\,dy$.

Board State & Physical Actions: Board State 6: The Orange Rule

To emphasize its importance for practical computation, the professor writes this finalized substitution rule on the board in bright orange chalk.

The Practical Substitution Rule

When substituting $x = \Psi(y)$, the differential transforms as:

$$\underbrace{dx}_{\text{Original Volume}} = \underbrace{|\det J\Psi(y)|}_{\text{Scaling Factor}} \underbrace{dy}_{\text{New Volume}}$$

Yielding the clean theorem:

$$\int_A f(x)\,dx = \int_{\Phi(A)} f(\Psi(y))\,|\det J\Psi(y)|\,dy$$

5 Example: Area of a Unit Disk & The Definition of π

Cleaned Spoken Protocol [Time: 00:12:15 - 00:15:00]

Let us compute a concrete example. But before we do, I must ask you: how was the constant π officially defined in your Analysis 1 course?
(A student answers correctly: "It is defined as the unique number in the open interval $(0, 4)$ such that the sine of π is zero.")
Exactly! You know many things about π from geometry, but your official, rigorous definition is that π is the first positive root of the sine function.
The sine and cosine functions themselves are defined purely analytically (usually via power series), and their defining properties are that the derivative of sine is cosine, the derivative of cosine is negative sine, $\sin(0) = 0$, and $\cos(0) = 1$. Those relations are definitive.

Board State & Physical Actions: Board State 7: Defining Pi rigorously

The professor steps to the far right board to lay down the absolute axioms of Analysis 1 before computing the area of a circle. He wants to prove the area is π *without* using circular geometric logic.

Rigorous Axioms from Analysis 1: The trigonometric functions are defined purely by their analytical properties (not triangles):

$$\sin'(x) = \cos(x) \tag{2}$$
$$\cos'(x) = -\sin(x) \tag{3}$$
$$\sin^2(x) + \cos^2(x) = 1 \tag{4}$$

The constant π is defined analytically as the unique root:

$$\sin(\pi) = 0 \quad \text{(for the first positive root)}$$

Redundant Explanation of Step: By establishing these purely algebraic definitions, the professor is ensuring that when we compute the integral of a disk and get the symbol π, we are arriving at it through rigorous calculus and limits, not by relying on a high-school geometric formula ($A = \pi r^2$) which assumes the conclusion we are trying to prove.

Insight: Avoiding Circular Logic

This is a brilliant teaching moment. It is very easy for students to take geometry for granted. By forcing them to recall that π is just "the root of a specific power series",

the professor elevates the upcoming Change of Variables computation from a "boring high-school geometry problem" into a "profound demonstration of analytical power."

Note: The video segment ends right as the professor finishes writing these trigonometric axioms on the board. In the next segment, he will define the polar coordinate mapping $\Psi(y_1, y_2)$ and use the Orange Substitution Rule to calculate the Jacobian determinant and prove the area of the unit disk is exactly π.

6 Hyper-Detailed Pedagogical Blueprint Real Analysis: Change of Variables & Rigorous Foundations (Part 2)

Director's Cut & AI Extraction Protocol
Lecture Segment: 15:00 – 32:00

4 Computing the Area of a Unit Disk

Cleaned Spoken Protocol [Time: 00:15:00 - 00:18:37]

Now that we have established the rigorous analytical definition of π, we will prove—perhaps for the first time in your mathematical lives—that the area of the unit disk is exactly π.
In our rigorous language, we are asking for the 2-dimensional Jordan measure of the set B_1, which is the set of all points (x_1, x_2) such that $x_1^2 + x_2^2 \leq 1$. To compute this measure, we will use polar coordinates.
Let the first parameter y_1 represent the radius, and the second parameter y_2 represent the angle. We define our coordinate mapping Ψ such that $x_1 = y_1 \cos(y_2)$ and $x_2 = y_1 \sin(y_2)$. Our parameter domain D will be the open rectangle where the radius y_1 goes from 0 to 1, and the angle y_2 goes from 0 to 2π.

Board State & Physical Actions: Board State 8: Setting up the Transformation
The professor writes down the set definition for the disk and formally defines the polar mapping Ψ alongside its parameter domain D.

Mathematical Setup (Stroke-by-Stroke): The Unit Disk in the original coordinates:

$$B_1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$$

The Polar Coordinate Mapping Ψ:

$$\begin{aligned} x_1 &= y_1 \cos(y_2) \\ x_2 &= y_1 \sin(y_2) \end{aligned}$$

The Parameter Domain D:

$$D = \underbrace{(0,1)}_{\text{Radius } y_1} \times \underbrace{(0,2\pi)}_{\text{Angle } y_2} \implies \Psi : D \to B_1$$

Redundant Explanation of Step: We are transitioning from Cartesian coordinates (x_1, x_2) to polar coordinates (y_1, y_2). The domain D is strictly an *open* rectangle. This strict open boundary guarantees that Ψ remains a proper diffeomorphism (bijective and smooth), as including $y_1 = 0$ (the origin) or $y_2 = 0, 2\pi$ (the overlapping seam) would break the bijection.

Cleaned Spoken Protocol [Time: 00:18:37 - 00:20:17]

Our first crucial observation is that the image of our parameter domain, $\Psi(D)$, is not entirely equal to the closed disk B_1. Because we are using the open interval $(0,1)$ for the radius and $(0,2\pi)$ for the angle, our mapping completely misses the origin and the line segment lying on the positive x-axis.
However, we can easily prove that this missing 1-dimensional line segment has a 2-dimensional

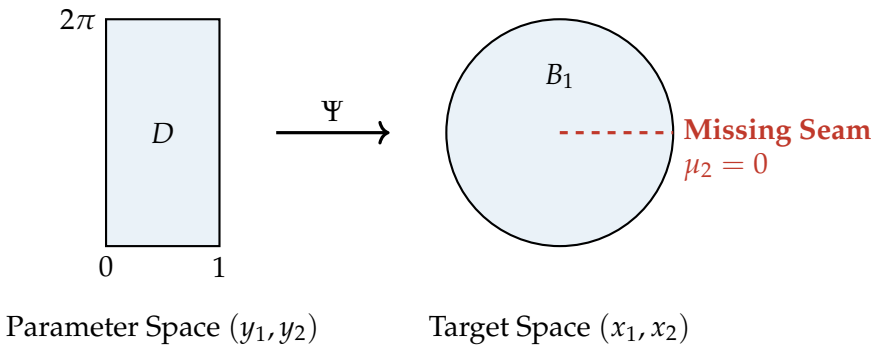
Jordan measure of zero. If you cover it with small squares of side length ϵ, the total area goes to zero. Therefore, even though the sets are not identical, their measures are perfectly equal: $\mu_2(B_1) = \mu_2(\Psi(D))$.

Insight: The "Measure Zero" Safety Net

This is a critical Real Analysis concept. Students often panic when a mapping isn't perfectly surjective over the boundaries. The professor is teaching them the "cheat code" of integration: integrals do not "see" sets of measure zero. You can freely delete boundaries, seams, and single points without changing the volume.

Board State & Physical Actions: TikZ: The Measure Zero Boundary Issue

The professor draws the mapping, actively highlighting the "missing" seam on the right side of the circle.



Cleaned Spoken Protocol [Time: 00:20:17 - 00:22:55]

Because the measures are equal, we can apply our Change of Variables substitution rule. The area of the disk is the integral of the constant function 1 over our domain D, multiplied by the determinant of the Jacobian matrix of Ψ.
Let us compute this Jacobian matrix. Taking the partial derivatives with respect to y_1 and y_2, we construct our 2×2 matrix. When we compute the determinant, we get $y_1 \cos^2(y_2) + y_1 \sin^2(y_2)$. Factoring out the y_1, we use our analytical axiom that $\cos^2 + \sin^2 = 1$. The determinant is simply y_1. Because y_1 is strictly positive on our domain, its absolute value is just y_1.

Board State & Physical Actions: Board State 9: The Jacobian Calculation

The professor explicitly writes out the matrix and its determinant. He jokes about making arithmetic mistakes, but executes it flawlessly.

Applying the Substitution Rule:

$$\mu_2(B_1) = \int_{\Psi(D)} 1\, dx = \int_D 1 \cdot \underbrace{|\det J\Psi(y_1, y_2)|}_{\text{Distortion Factor}}\, dy$$

Computing the Jacobian Matrix:

$$J\Psi(y_1, y_2) = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{pmatrix} = \begin{pmatrix} \cos(y_2) & -y_1 \sin(y_2) \\ \sin(y_2) & y_1 \cos(y_2) \end{pmatrix}$$

Evaluating the Determinant:

$$\begin{aligned}\det J\Psi &= \big(\cos(y_2)\cdot y_1\cos(y_2)\big) - \big(-y_1\sin(y_2)\cdot\sin(y_2)\big)\\ &= y_1\cos^2(y_2) + y_1\sin^2(y_2)\\ &= y_1\underbrace{\big(\cos^2(y_2)+\sin^2(y_2)\big)}_{=1\text{ (Axiom)}}\\ &= y_1\end{aligned}$$

Redundant Explanation of Step: The Jacobian determinant tells us exactly how much a tiny square of parameter space dy_1dy_2 is stretched when it is mapped into the disk. Because the determinant is y_1, it means the "stretching" scales linearly with the radius. Grids near the outer edge of the circle are stretched much larger than grids near the origin.

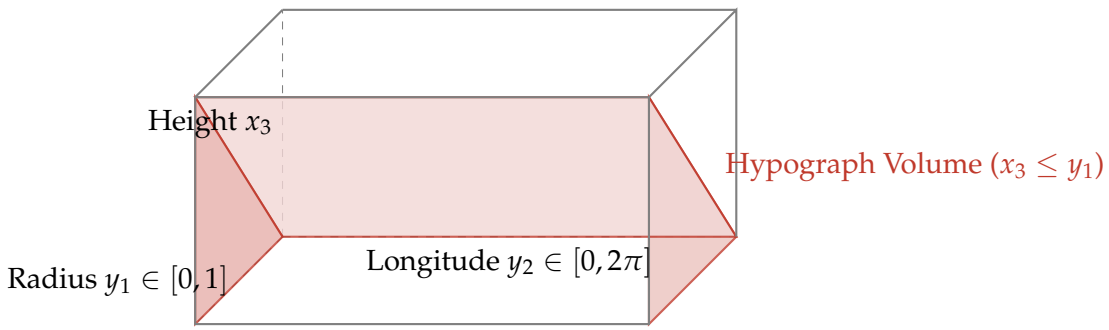
> ***Cleaned Spoken Protocol [Time: 00:22:55 - 00:26:33]***
>
> *Substituting this back in, we must compute the double integral of y_1 over our rectangular domain D. But we have a problem: we haven't learned how to systematically compute multivariable integrals yet!*
> *We can solve this using a geometric trick. The integral of a positive function over a 2D domain represents the 3-dimensional volume of its hypograph—the physical region trapped underneath the function's surface.*
> *Let's add a vertical coordinate, x_3. We want the volume where $x_3 \leq y_1$, over our base rectangle $(0,1)\times(0,2\pi)$. Consider a full 3D bounding box with dimensions $1\times 2\pi\times 1$. The total volume is 2π. Our function, $x_3 = y_1$, forms a slanted plane that cuts this box perfectly in half along its diagonal. Therefore, the volume of our hypograph is exactly half the volume of the box, which gives us π.*

Board State & Physical Actions: Board State 10: The Hypograph Wedge

The professor completes the proof not with calculus, but by drawing a 3D bounding box and showing how the function $f(y_1,y_2) = y_1$ cuts it precisely in half.

The Final Integration via Geometry:

$$\mu_2(B_1) = \int_{(0,1)\times(0,2\pi)} \underbrace{y_1}_{\text{Height } x_3}\ \underbrace{dy_1\,dy_2}_{\text{Base Area}} \tag{5}$$



Redundant Explanation of Step: By interpreting the integral physically, we see the function y_1 creates a wedge. Because the wedge is a perfect diagonal bisection of a rectangular prism, we can compute its volume using simple geometry instead of advanced integration rules.

$$\mu_2(B_1) = \frac{1}{2}\,\text{Vol}(\text{Bounding Box}) = \frac{1}{2}(1\cdot 2\pi\cdot 1) = \pi$$

5 Example: Volume of the Unit Ball in $\mathbb{R}^3$

Cleaned Spoken Protocol [Time: 00:26:33 - 00:32:00]

That trick is wonderful, but let us move to a harder question. Let's compute the volume of the unit ball in $\mathbb{R}^3$. We know from geometry that the answer should be $\frac{4}{3}\pi$, but we have not proved it analytically yet.
We will try the exact same approach using 3D polar coordinates—spherical coordinates. We like spherical coordinates because they flawlessly transform rectangular boxes into spheres. Let our mapping be $\Psi(y_1, y_2, y_3)$.
The first coordinate, y_1, is always the radius. The second coordinate, y_2, is the longitude—the deviation from a fixed meridian. The third coordinate, y_3, represents the elevation. Let me write the formulas down...

Board State & Physical Actions: Board State 11: The Famous Mistake

The professor attempts to write the spherical coordinates. He intends for y_3 to be the "elevation" (latitude, measured from the equator). However, muscle memory kicks in, and he writes the standard physics convention where the angle is measured from the North Pole (colatitude). This is mathematically incorrect for his stated domain.

The Initial (Incorrect) Board State:

$$\begin{aligned} x_1 &= y_1 \cos(y_2) \sin(y_3) \\ x_2 &= y_1 \sin(y_2) \sin(y_3) \\ x_3 &= y_1 \cos(y_3) \end{aligned} \tag{6}$$

Insight: The Coordinate Trap

There are two competing standards for spherical coordinates. Mathematics often uses latitude (angle from the equator, meaning $z = r \sin \phi$). Physics usually uses colatitude (angle from the North Pole, meaning $z = r \cos \phi$). The professor mixed the words of one with the math of the other. This happens to the best of us, and it is a phenomenal active-learning moment for the room.

Cleaned Spoken Protocol [Time: 00:32:00 - 00:32:53]

Wait, looking at this... did I make a mistake?
Student: "If we would like y_3 to represent the elevation from the equator, shouldn't we switch the sine and cosine in the formula for the polar coordinates?"
Ah! The y_3 is the elevation, right? Yes, you are completely right. If we use the cosine for the x_3 coordinate, we are measuring the deviation from the North Pole. If we want the elevation from the equator, the vertical x_3 coordinate must be governed by the sine function. Let me fix this on the board. The x_3 gets the sine, and the others get the cosine. Excellent catch.

Board State & Physical Actions: Board State 12: The Correction

The professor erases the sines and cosines and formally swaps them, bringing the math into perfect alignment with his intended domain for elevation.

The Corrected Board State (Stroke-by-Stroke): The mapping $\Psi : D \to \mathbb{R}^3$ for elevation-

based spherical coordinates:

$$
\begin{aligned}
x_1 &= y_1 \cos(y_2) \underbrace{\cos(y_3)}_{\text{Corrected}} \\
x_2 &= y_1 \sin(y_2) \underbrace{\cos(y_3)}_{\text{Corrected}} \\
x_3 &= y_1 \underbrace{\sin(y_3)}_{\text{Corrected}}
\end{aligned}
\tag{7}
$$

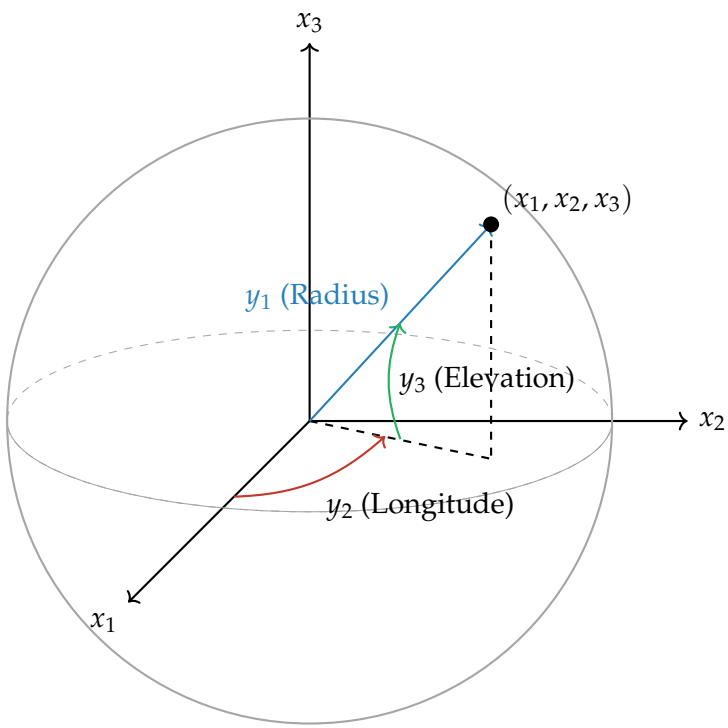
Redundant Explanation of Step: By setting $x_3 = y_1 \sin(y_3)$, we ensure that when the elevation angle $y_3 = 0$, our height x_3 is 0 (putting us on the equator). When $y_3 = \pi/2$, $\sin(\pi/2) = 1$, putting us exactly at the North Pole $x_3 = y_1$.

The parameter domain D is formally defined as:

$$
D = \underbrace{(0,1)}_{y_1(\text{Radius})} \times \underbrace{(0,2\pi)}_{y_2(\text{Longitude})} \times \underbrace{\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)}_{y_3(\text{Elevation})}
$$

Board State & Physical Actions: TikZ: Spherical Coordinates Visualized

Visual representation of the corrected coordinate system, explicitly showing y_3 sweeping up from the equatorial plane.



Board State & Physical Actions: End of Second Segment

The lecture protocol currently ends here. The next mathematical step will be computing the 3×3 Jacobian determinant of this corrected system, leading directly into Cavalieri's Principle to evaluate the resulting 4-dimensional integral.

6 Hyper-Detailed Pedagogical Blueprint Real Analysis: Change of Variables & Rigorous Foundations (Part 3)

Director's Cut & AI Extraction Protocol
Lecture Segment: 32:00 – End

5 Completing the 3D Unit Ball Setup

Cleaned Spoken Protocol [Time: 00:32:53 - 00:36:10]

Now that we have the correct spherical coordinates defined, we need the Jacobian determinant. I am not going to spend seven minutes on the board computing this 3×3 matrix, but I highly encourage you to do it at home. Because of the trigonometric identity $\sin^2(\theta) + \cos^2(\theta) = 1$, many terms simplify beautifully.
In the end, the determinant of the Jacobian evaluates to $y_1^2 \cos(y_3)$. Notice that because our elevation angle y_3 goes strictly from $-\pi/2$ to $\pi/2$, the cosine function is always positive. Therefore, the absolute value of the determinant is simply $y_1^2 \cos(y_3)$ itself.
Using our Change of Variables formula, the volume of the unit ball in $\mathbb{R}^3$ is equal to the integral over our rectangular parameter domain D.

Board State & Physical Actions: Board State 13: Finalizing the 3D Integral
The professor skips the brute-force matrix algebra, writes the final Jacobian determinant, and constructs the triple integral on the board.

Mathematical Setup (Stroke-by-Stroke): The Jacobian determinant simplifies to:

$$\det J\Psi = y_1^2 \cos(y_3)$$

Because $y_3 \in (-\pi/2, \pi/2)$, $\cos(y_3) > 0$. Applying the substitution rule:

$$\mu_3(B_1) = \int_D \underbrace{y_1^2 \cos(y_3)}_{\text{Distortion Factor}} \, dy = \int_0^1 \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} y_1^2 \cos(y_3) \, dy_3 \, dy_2 \, dy_1 \tag{8}$$

Cleaned Spoken Protocol [Time: 00:36:10 - 00:40:07]

Now, you might ask me: "How do we do a geometric trick to compute this, like we did with the 2D wedge?" The answer is, we don't. I know of no simple geometric trick to evaluate this 4-dimensional hyper-volume.
While we could try to invent more sophisticated coordinate changes, we will instead approach this integral systematically. To compute integrals of this type effortlessly, we must introduce the slicing idea. We will establish Cavalieri's Principle, which leads directly to Fubini's Theorem.

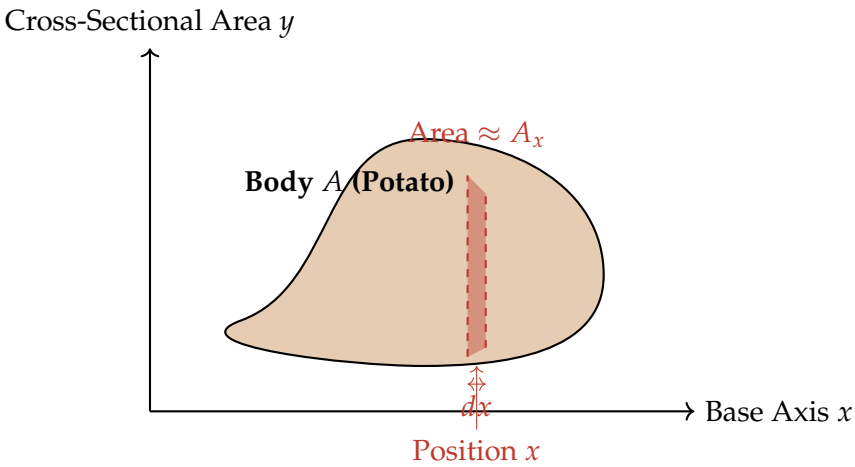
6 The Slicing Idea: Cavalieri's Principle

Cleaned Spoken Protocol [Time: 00:40:07 - 00:42:00]

Let me motivate the slicing idea informally. The Cavalieri Principle states that if you have a 3D body—which for us will be a Jordan measurable set—you can compute its total volume by slicing it.
Imagine a potato. If you take a potato and cut a very thin vertical slice, that slice is almost perfectly uniform from its front face to its back face. You can approximate the volume of that single slice by multiplying its cross-sectional area by its tiny thickness. If you sum the volumes of all these slices, the total volume of the potato is perfectly preserved.

Board State & Physical Actions: TikZ: The Potato Analogy

The professor draws a 3D blob (a potato) and highlights a single, thin vertical slice to illustrate the concept of $V \approx \sum A(x)\Delta x$.



Cleaned Spoken Protocol [Time: 00:42:00 - 00:46:30]

How do we do this systematically? We state it in terms of Jordan measures. We split our n-dimensional space, $\mathbb{R}^n$, into a product space: $\mathbb{R}^k \times \mathbb{R}^{n-k}$. Let the base coordinates be x and the vertical coordinates be y.
Let $A \subset \mathbb{R}^n$ be a Jordan measurable set. For a specific base point x, we define the slice A_x as the set of all vertical points y that lie inside A.
*Here is the technical hurdle: Just because A is Jordan measurable in $\mathbb{R}^n$ does not guarantee that the slice A_x is Jordan measurable in the lower-dimensional space $\mathbb{R}^{n-k}$. But we don't care! The outer measure and inner measure are *always* well-defined for any bounded set. So, we define the upper marginal density $\overline{\Phi}(x)$ using the outer measure, and the lower marginal density $\underline{\Phi}(x)$ using the inner measure.*

Board State & Physical Actions: Board State 14: Formalizing the Slices

The professor writes down the formal set notation for slices and explicitly addresses the "non-measurable slice" edge case using inner/outer measures.

Mathematical Setup (Stroke-by-Stroke): Split the space: $\mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^{n-k}$. A point is (x, y). Let $A \subset \mathbb{R}^n$ be Jordan measurable.

Definition 1 (Slices)**.** For a fixed base coordinate $x \in \mathbb{R}^k$, the slice A_x is defined as:

$$A_x = \{y \in \mathbb{R}^{n-k} \mid (x, y) \in A\}$$

To bypass slices that are not Jordan measurable, we define the **marginal densities**:

$$\overline{\Phi}(x) = \mu^{\text{out}}_{n-k}(A_x) \quad \text{(Upper Marginal)}$$
$$\underline{\Phi}(x) = \mu^{\text{in}}_{n-k}(A_x) \quad \text{(Lower Marginal)}$$

Cavalieri's Principle

Both marginal functions are Riemann integrable over x, and their integrals yield the total volume of A:

$$\int \overline{\Phi}(x)\, dx = \int \underline{\Phi}(x)\, dx = \mu_n(A)$$

Insight: The "I Don't Care" Philosophy

When the professor says "I don't care" about the slice not being Jordan measurable, he is imparting profound mathematical wisdom. In integration, sets of measure zero or pathological boundary anomalies disappear when you sum over them. The upper and lower integrals act as a squeeze theorem—even if individual slices are "fuzzy," integrating over the entire domain squeezes that fuzziness down to zero.

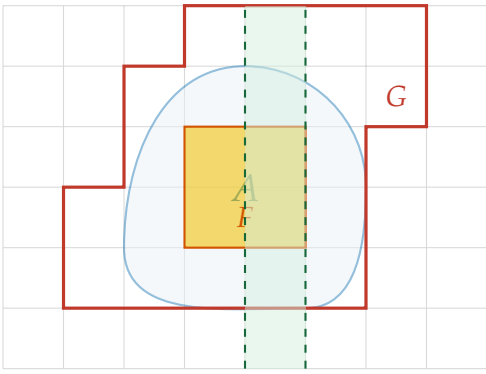
7 Proof by Counting Pixels

Cleaned Spoken Protocol [Time: 00:46:30 - 00:52:00]

How do we prove this? Because A is Jordan measurable, we can approximate it using a very small grid of dyadic cubes—like pixels on a screen.
For any $\epsilon > 0$, we can find an inner approximation F (made entirely of yellow pixels inside A) and an outer approximation G (made of red pixels covering A), such that the volume difference between G and F is less than ϵ.
Because these sets are built from perfect cubes aligned to a grid, what happens when we slice them? As long as our base point x stays within the same column of the grid, the slice of F or G cannot change! Therefore, the measures of these slices, $\mu(F_x)$ and $\mu(G_x)$, are just dyadic step functions.

Board State & Physical Actions: TikZ: The Dyadic Pixel Proof

The professor illustrates the inner (F) and outer (G) approximations. He notes that integrating a dyadic step function is just combinatorics—counting the total number of pixels.



Slice Column x

Proof Logic (Redundant Mathematical Breakdown): 1. A is sandwiched: $F \subset A \subset G$. 2. This forces the slices to be sandwiched: $F_x \subset A_x \subset G_x$. 3. Taking the measure of the slices:

$$\mu(F_x) \leq \underline{\Phi}(x) \leq \overline{\Phi}(x) \leq \mu(G_x)$$

4. Integrating the step functions $\mu(F_x)$ and $\mu(G_x)$ over the base dx simply counts the total number of pixels in F and G. Thus:

$$\text{Vol}(F) \leq \int \underline{\Phi}(x)dx \leq \int \overline{\Phi}(x)dx \leq \text{Vol}(G)$$

5. Since we chose our grid such that $\text{Vol}(G) - \text{Vol}(F) < \epsilon$, as $\epsilon \to 0$, the outer and inner pixel volumes squeeze the marginal integrals perfectly to $\text{Vol}(A)$.

8 Fubini's Theorem: Cavalieri for Functions

Cleaned Spoken Protocol [Time: End of Lecture Segment]

Fubini's theorem is the immediate consequence of Cavalieri's Principle applied to functions instead of just shapes.
If you have a function $f(x, y)$ that is Riemann integrable over A, you can compute the total multi-dimensional integral by integrating over the slices first. You integrate over y for a fixed x, and then you sum those results over all x.
The proof is identical. In fact, remember that the integral of a function is just the volume of its hypograph! Fubini's Theorem is quite literally just Cavalieri's Principle applied to the $n + 1$ dimensional hypograph wedge of the function.

Fubini's Theorem

$$\int_A f(x, y)\, d(x, y) = \int_{\mathbb{R}^k} \left(\int_{\mathbb{R}^{n-k}} \tilde{f}_x(y)\, dy \right) dx$$

Insight: The Grand Finale

The entire lecture culminates here. The 2D disk trick (where the function $x_3 = y_1$ created a wedge) was a manual, geometric application of Fubini's theorem! The professor used a visual trick to solve the integral before proving the theorem that justifies it, completing the pedagogical loop.

9 Monday, April 15th: Second Half of the Lecture Cavalieri's Principle: Statement and Proof via Dyadic Cubes

AI Extraction Protocol
Segment: 00:00 – 15:15

Previous Board State

As the video begins, there is a pre-existing formula on the left blackboard from the previous session: $\mu_3(B_1^3) = \mu_3(\Psi(D)) = \int_{(0,1)\times(0,2\pi)\times(\frac{\pi}{2},\pi)} y_1^2 \cos y_3 \, dy = \dots$ with the words "Slicing idea" written above it. On the right board, the heading "Cavalieri principle" is written alongside a 3D sketch of a body $A \subset \mathbb{R}^3$ being sliced into parallel disks, with the formula: $\text{volume}(A) = \sum_{\text{all slices}} \cdots$

9.1 Geometric Setup of Cavalieri's Principle

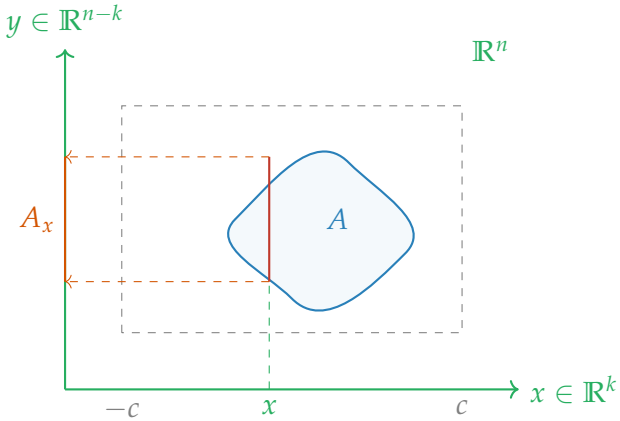
Cleaned Spoken Protocol [Time: 00:00:08 - 00:02:20]

Let me continue with the statement. We are trying to formalize this theorem that I informally described as "cutting the potato." But now we are trying to be more serious, and I will introduce the proper notations.
This theorem, you will see, is a bit the opposite of the Change of Variables. It is a very simple statement, but it has a very hard proof. The setup involves very complicated statements, but the proof itself is actually very easy, right?
Let me do a drawing of the measure space. We have our set A, right, that is contained in a box. Here we will put the X-axis and the Y-axis. The X-axis actually represents $\mathbb{R}^k$, and the Y-axis represents $\mathbb{R}^{n-k}$. The whole space is $\mathbb{R}^n$, which is the product of the two.
Okay, now we are saying that our set is contained in some bounding box. It is a very large box, going from $-c$ to c. Now, let's look inside this box. If we pick some point x here on the X-axis, I will consider the slice. The relevant piece of the set is this vertical slice over here, right? This is what I am calling A_x. The set A_x will be the projection of this slice onto the vertical axis.

Board State & Physical Actions: Board State 1: The Potato and the Slice

The professor moves to a clean section of the left board. He draws a large, smooth, blob-like shape (the "potato") and labels it A. He draws the Cartesian coordinate axes, explicitly defining the X-axis as $\mathbb{R}^k$ and the Y-axis as $\mathbb{R}^{n-k}$. He draws a dashed bounding box around A. He then picks a specific point x on the horizontal axis, draws a vertical line segment exactly through A, and projects that 1D slice onto the Y-axis, labeling it A_x.

Geometric Visualization Setup



Cleaned Spoken Protocol [Time: 00:02:20 - 00:04:50]

So you take this set and you project what? You take the y-coordinates. Given the x, you take the y-coordinates of the pairs (x, y) that belong to A. Since x is fixed, what you are looking at is: you go to x, you look at which y-coordinates you have inside the set, and this is the set A_x.
Now, this set here is very nice because in the drawing it is just a 1D interval, right? But we might be in very high dimensions. So A_x is just a subset of $\mathbb{R}^{n-k}$ a priori, and it is a bounded set. We know that it is strictly contained within $(-c, c)^{n-k}$.
Because we know the entire set A is Jordan measurable, we might hope the slice is too. But maybe for some pathological slices, the set is not measurable because the outer and inner measures do not coincide. However, we can surely consider the marginals! We can always measure the outer measure $\mu_{n-k,out}$ and the inner measure $\mu_{n-k,in}$.
The measure from the inside and the outside of the set is always well-defined, and this defines the functions $\overline{\varphi}(x)$ and $\underline{\varphi}(x)$, right? So this is the setup.

Board State & Physical Actions: Board State 2: Formalizing the Slices

The professor moves to the right board to formalize the geometric picture into strict mathematical notation. He writes the definition of the slice A_x and the upper/lower marginal densities.

Formal Definition of the Slice and Marginals

Let $\mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^{n-k}$, with coordinates $(x, y) \in \mathbb{R}^n$.
Let $A \subset \mathbb{R}^n$ be Jordan measurable, with $A \subset (-c, c)^n$.
Define the slices: for $x \in (-c, c)^k$,

$$A_x = \{y \in \mathbb{R}^{n-k} \mid (x, y) \in A\} \subset (-c, c)^{n-k}$$

We define the upper and lower marginal densities $\overline{\varphi}, \underline{\varphi} : [-c, c]^k \to \mathbb{R}$ as:

$$\overline{\varphi}(x) = \mu_{n-k,\text{out}}(A_x)$$
$$\underline{\varphi}(x) = \mu_{n-k,\text{in}}(A_x)$$

Inner and Outer Guarantees

Even if a solid body A is perfectly measurable, a specific 1D slice could theoretically intersect a fractal boundary or a cloud of points, making that exact slice non-measurable. To bypass this entirely, the professor rigorously defines the marginals using the universally safe inner and outer measures.

10 The Theorem Statement

Cleaned Spoken Protocol [Time: 00:04:50 - 00:07:30]

And the theorem says that under these assumptions, both functions $\overline{\varphi}$ and $\underline{\varphi}$ mapping from $[-c, c]^k \to \mathbb{R}$ are Riemann integrable.
And the integral of them over the set $[-c, c]^k$—the integral of $\underline{\varphi}(x)\,dx$ is equal to the integral of $\overline{\varphi}(x)\,dx$, and both are equal to the measure of A, $\mu_n(A)$.
So this integral is just you summing up the $\overline{\varphi}$. The $\overline{\varphi}$ is the measure of each slice! Now, you can tell me, "Maybe for some slices this measure is not well-defined because the outer and the inner measure do not coincide!" I tell you: do not worry! Just put either the outer or the inner. I don't care, because when you integrate the outer, or the inner, or anything in between, you will always recover the exact measure of your set A.
Okay, let's prove that. To do the proof, I will do another drawing.

Cavalieri's Principle

If $A \subset \mathbb{R}^n$ is Jordan measurable, then the marginal functions $\overline{\varphi}, \underline{\varphi} : [-c, c]^k \to \mathbb{R}$ are Riemann integrable, and:

$$\int_{[-c,c]^k} \underline{\varphi}(x)\,dx = \int_{[-c,c]^k} \overline{\varphi}(x)\,dx = \mu_n(A)$$

11 The Proof Setup: Dyadic Cubes

Cleaned Spoken Protocol [Time: 00:07:30 - 00:11:00]

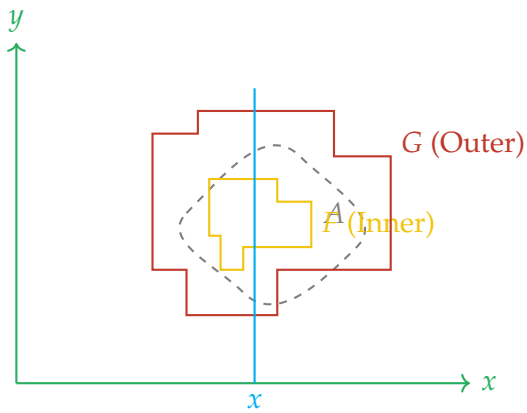
Always when we want to prove something about Jordan measures, remember that we are approximating our set from the inside and the outside. We will take an approximation from the inside, right? I'm approximating from the inside, and then there will be the approximation from the outside.
Let me describe a bit with a picture of what we are going to do in the proof, and later we will do the symbols. Since our set is Jordan measurable, we can take a small pixel size of 2^{-p}. We can take an inner and an outer approximation over our set by dyadic cubes. So by pixels! This is the inner, this is the outer.
The difference between these two is less than epsilon. We can make the volume of the red outer approximation minus the volume of the yellow inner approximation less than epsilon for whatever precision we want.
Now we will see what happens when we take a slice of this pixelated set. Let's say we take x and we slice, right? What we see is that the slice of A contains the slice of the yellow, which is contained in the slice of the red. And when you start to move x, whenever you move x less than

the size of the pixel, you are not jumping to a new cube!

Board State & Physical Actions: Board State 3: The Pixelated Sandwich

The professor moves to a new board. He redraws the smooth blob A. Using yellow chalk, he meticulously draws a jagged, pixelated boundary entirely inside A (the inner approximation F). Using red chalk, he draws a larger pixelated boundary entirely enclosing A (the outer approximation G). He draws a vertical line to show how the 1D slice of F is a subset of the slice of A, which is a subset of the slice of G.

Geometric Proof Setup: Inner and Outer Approximations



Cleaned Spoken Protocol [Time: 00:11:00 - 00:15:15]

Because you are not jumping to a new cube, the marginal densities computed for the inner and outer approximations are dyadic step functions!
This is the key observation. They are dyadic step functions in x, and they will approximate the Riemann integral. The dyadic step functions are what we use to approximate the integral.
So this is how we will try to proceed. Let's write this formally. The idea is that given a dyadic cube—I will take one of size 2^{-p} in $\mathbb{R}^n$. Let me call this $Q_p^n(z)$. So $Q_p^n(z) = z + 2^{-p}[0,1)^n$, right? Since there will be cubes of several dimensions, let me put the dimension here as an index n.
Now, if z is of the form (a,b) where $a \in \mathbb{R}^k$ and $b \in \mathbb{R}^{n-k}$—meaning in this splitting I take a z that has coordinates a and b—then this n-dimensional cube $Q_p^n(z)$ is equal to $Q_p^k(a) \times Q_p^{n-k}(b)$, right?
So I am saying something very complicated to write, but very trivial geometrically. On the picture, I am saying that the n-dimensional cube over here is always the product of a base cube over here times the vertical cube over there!

Board State & Physical Actions: Board State 4: Formalizing the Dyadic Cubes

The professor moves to the far right board to set up the rigorous notation for the dyadic cubes (the "pixels"). He shows how an n-dimensional cube factors smoothly into a Cartesian product of a k-dimensional cube and an $(n-k)$-dimensional cube, matching the X and Y axis splitting perfectly.

Dyadic Cube Definitions

A dyadic cube of side length 2^{-p} in $\mathbb{R}^n$ anchored at z is defined as:

$$Q_p^n(z) = z + 2^{-p}[0,1)^n$$

If the anchor point splits across our axes such that $z = (a, b)$ where $a \in \mathbb{R}^k$ and $b \in \mathbb{R}^{n-k}$, then the n-dimensional cube factors across the dimensions:

$$Q_p^n(z) = Q_p^k(a) \times Q_p^{n-k}(b)$$

Where, generally for any dimension m:

$$Q_p^m(i) = i + 2^{-p}[0,1)^m \subset \mathbb{R}^m$$

Insight: Calculus via Combinatorics

By replacing the continuous, smooth curve with a grid of literal squares (dyadic cubes), the nasty continuous calculus problem drops away. As long as your x-coordinate stays within the width of a single vertical column of squares, the vertical slice does not change its length. It is piecewise constant. A terrifying integration problem becomes the simple integration of a step function!

12 Cavalieri's Principle: Integration of Step Functions and The Squeeze

AI Extraction Protocol
Segment: 15:15 – 30:42

Segment Continuation

Continuing directly from the previous segment, the professor has just established that the n-dimensional set A can be approximated from the inside and the outside using a discrete grid of dyadic cubes ("pixels"). He now transitions to the algebraic formalization of these geometric bounds.

13 Formalizing the Approximations

Cleaned Spoken Protocol [Time: 15:15 - 18:22]

Let us formalize these geometric bounds. Given any arbitrary precision $\varepsilon > 0$, we can choose a sufficiently fine pixel size—let us say, governed by some integer $p \in \mathbb{N}$ determining the side length 2^{-p}. By the very definition of Jordan measurability, there must exist an inner dyadic set F and an outer dyadic set G such that our set A is completely sandwiched between them.
Furthermore, we are guaranteed that the total volume of the large outer approximation minus the total volume of the small inner approximation is strictly less than ε.
Now, let us examine the vertical cross-sections of these pixelated approximations. For any base coordinate x residing within our bounding box $[-c, c]^k$, we define the slice of the outer approximation, G_x, as the set of all vertical y-coordinates such that the composite point (x, y) belongs to G. We define the slice of the inner approximation, F_x, in the exact same manner. Recall that both F and G are simply finite unions of dyadic cubes of pixel size 2^{-p}.

Board State & Physical Actions: Board State 6: Establishing Jordan Bounds

The professor moves to a clean section of the center board to begin the formal algebraic proof. He writes down the standard definition of a Jordan measurable set using the dyadic approximations F and G, followed by the set-builder notation for their respective vertical slices.

Inner and Outer Dyadic Sets and Slices

Given $\varepsilon > 0$, $\exists p \in \mathbb{N}$ and dyadic sets F, G (composed of cubes of side length 2^{-p}) such that:

$$F \subset A \subset G \quad \text{and} \quad \mu_n(G) - \mu_n(F) < \varepsilon$$

For all $x \in [-c, c]^k$, we define the vertical slices:

$$G_x = \{y \in \mathbb{R}^{n-k} \mid (x, y) \in G\}$$
$$F_x = \{y \in \mathbb{R}^{n-k} \mid (x, y) \in F\}$$

14 Pixels to Step Functions

Cleaned Spoken Protocol [Time: 18:22 - 21:50]

Let us define a new function, $f(x)$, which represents the $(n-k)$-dimensional volume of the inner slice F_x. Similarly, let $g(x)$ represent the volume of the outer slice G_x.
Because these sets are constructed entirely from a uniform grid of dyadic cubes, the geometry of the vertical slice does not change continuously as you move x. As long as your coordinate x remains within the boundaries of a single base pixel $Q_p^k(a)$, the volume of the vertical slice intersecting it remains perfectly constant.
What does it mean for a function to be piecewise constant across a grid of cubes? It means that $f(x)$ and $g(x)$ are, by definition, dyadic step functions. Additionally, since our entire set is contained within the bounding box, these functions trivially evaluate to zero outside of $[-c,c]^k$. Consequently, we have successfully bounded our continuous problem with adequate step functions, which are exactly the tools we need to approximate the Riemann integral.

Insight: The Geometric Leap

The professor explicitly bridges the physical drawing of "pixels" to the algebraic definition of a step function. Because F is constructed purely of vertical columns of uniform width 2^{-p}, moving x horizontally within the boundary of a single column does not change the vertical cross-section F_x. Thus, the volume function $f(x)$ is perfectly flat (constant) across that entire base pixel.

Step Function Property

Define the slice volume functions $f, g : [-c,c]^k \to \mathbb{R}$ as:

$$f(x) := \mu_{n-k}(F_x)$$
$$g(x) := \mu_{n-k}(G_x)$$

Because F and G are finite unions of dyadic cubes, $f(x)$ and $g(x)$ are constant on every base cube $Q_p^k(a)$. Furthermore, $f(x) = g(x) = 0$ outside of $[-c,c]^k$.
$\implies$ f and g are **dyadic step functions**.

15 The Combinatorial Counting Argument

Cleaned Spoken Protocol [Time: 21:50 - 25:30]

Now, let us integrate these step functions. What is the integral of $f(x)\,dx$ over our base space? Because we are operating in the pixelated world of dyadic cubes, integrating the volume of the slices over the base domain degenerates into a simple counting argument! You are merely looking at each column and asking, "How many pixels are stacked here?" When you integrate, you are simply summing those discrete columns together to recover the total number of pixels in the entire shape.
There are no limits here, and no continuous approximation errors. It is pure combinatorics. Therefore, the integral of our inner step function $f(x)$ evaluates exactly to the total n-dimensional volume of the inner approximation, $\mu_n(F)$. By identical reasoning, the integral of $g(x)$ evaluates exactly to $\mu_n(G)$.

Board State & Physical Actions: Board State 7: Integrating the Step Functions

He writes the integral of $f(x)$ and $g(x)$ across the base space. He explicitly equates the integral over the base space to the total volumetric measure of the multi-dimensional approximation sets.

Combinatorial Integration

Because f and g are dyadic step functions, their Riemann integrals are precisely the sum of the volumes of their constituent n-dimensional cubes:

$$\int_{[-c,c]^k} f(x)\,dx = \mu_n(F)$$
$$\int_{[-c,c]^k} g(x)\,dx = \mu_n(G)$$

Calculus Becomes Counting

This is a beautiful pedagogical moment. The professor demystifies multi-dimensional integration by showing that when dealing with step functions constructed from a uniform grid, the Riemann integral becomes simple addition. You take the volume of a 2D slice, multiply it by the tiny 1D width of the base pixel (dx), and you get the 3D volume of that column. Summing them all up gives the total volume.

16 The Chain of Inequalities

Cleaned Spoken Protocol [Time: 25:30 - 27:30]

We now rely on our fundamental geometric inclusion. Because the inner approximation F is contained within A, which is in turn contained within G, this strict nesting applies to every single vertical slice. Thus, $F_x \subset A_x \subset G_x$.
This geometric nesting forces a strict ordering on the measures of these slices. The measure of the inner slice, $f(x)$, must be less than or equal to the inner measure of A_x, which is our lower marginal $\underline{\varphi}(x)$. In turn, the outer measure of A_x, which is our upper marginal $\overline{\varphi}(x)$, must be less than or equal to the measure of the outer slice, $g(x)$.
We have successfully constructed a chain of inequalities for our functions. A foundational property of Riemann integration is that if functions are strictly ordered, their integrals maintain that exact order. We will apply the lower Riemann integral to the lower marginal, and the upper Riemann integral to the upper marginal, preserving the entire inequality chain from end to end.

Board State & Physical Actions: Board State 8: Establishing the Squeeze

The professor moves back to the center board. He uses the geometric inclusion to establish an ordered chain of marginal densities. He then applies the lower and upper Riemann integrals to this chain, drawing a long series of inequalities.

Squeezing the Marginals

From the inclusion $F \subset A \subset G$, we have for all slices x:

$$F_x \subset A_x \subset G_x$$

Taking the respective measures preserves the ordering:

$$\underbrace{\mu_{n-k}(F_x)}_{f(x)} \leq \underbrace{\mu_{n-k,\text{in}}(A_x)}_{\underline{\varphi}(x)} \leq \underbrace{\mu_{n-k,\text{out}}(A_x)}_{\overline{\varphi}(x)} \leq \underbrace{\mu_{n-k}(G_x)}_{g(x)}$$

Integrating across $[-c, c]^k$ preserves the inequality chain:

$$\int f(x)\,dx \leq \underline{\int} \underline{\varphi}(x)\,dx \leq \overline{\int} \overline{\varphi}(x)\,dx \leq \int g(x)\,dx$$

17 Conclusion of the Proof

Cleaned Spoken Protocol [Time: 27:30 - 30:42]

Let us substitute the combinatorial volumes we found earlier into the bounding ends of this inequality. We know that the integral of $f(x)$ is exactly $\mu_n(F)$, and the integral of $g(x)$ is exactly $\mu_n(G)$.
Therefore, our unknown marginal integrals are permanently trapped between the volume of the inner approximation and the volume of the outer approximation. But remember our initial premise! We constructed our pixels specifically such that the difference between the large outer volume and the small inner volume is strictly less than ε.
What happens as we send ε to zero? The outer and inner bounds squeeze tightly together. Because the marginal integrals are trapped in the middle, and because they are entirely independent of our arbitrary choice of ε, they are forced to equal the exact same value.
This simultaneously proves two things: first, because the lower and upper integrals coincide, the marginal functions are definitively Riemann integrable. Second, because they are squeezed between bounds that converge to the true volume of the shape, their integral evaluates exactly to $\mu_n(A)$. And that completes the proof of Cavalieri's Principle.

Board State & Physical Actions: Board State 9: Substituting and Concluding

He substitutes $\mu_n(F)$ and $\mu_n(G)$ into the bounding ends of the integral inequality chain. He circles the ε condition, shows the bounds collapsing, and writes the final conclusion, boxing the result with a triumphant Q.E.D. mark.

The Final Inequality Chain

Substituting the known combinatorial integrals:

$$\mu_n(F) \leq \underline{\int} \underline{\varphi}(x)\,dx \leq \overline{\int} \overline{\varphi}(x)\,dx \leq \mu_n(G)$$

We know by our initial choice of dyadic approximations that:

$$\mu_n(G) - \mu_n(F) < \varepsilon$$

Conclusion of Cavalieri's Principle

Since the upper and lower integrals of the marginal functions are squeezed between $\mu_n(F)$ and $\mu_n(G)$ for any $\varepsilon > 0$, and the length of that interval vanishes as $\varepsilon \to 0$, we conclude:

$$\int_{[-c,c]^k} \underline{\varphi}(x)\, dx = \int_{[-c,c]^k} \overline{\varphi}(x)\, dx = \mu_n(A)$$

Therefore, $\underline{\varphi}$ and $\overline{\varphi}$ are Riemann integrable, completing the proof. ■

End of Segment

The lecture video segment concludes precisely here at 30:42 with the completion of Cavalieri's Principle for sets.

18 From Cavalieri to Fubini: Integrating Functions via Hypographs

AI Extraction Protocol
Segment: 30:42 – 45:00 (End of Lecture)

Segment Transition

The professor has just finished the rigorous dyadic-cube proof of Cavalieri's Principle for n-dimensional sets. He erases the center and right chalkboards to transition to the ultimate goal of the lecture: applying this geometric slicing principle to the calculus of multi-variable functions.

19 From Sets to Functions: The Hypograph

Cleaned Spoken Protocol [Time: 30:42 - 34:15]

We have successfully and rigorously established Cavalieri's Principle for the volumes of multi-dimensional sets. But we are taking a class on Analysis, not just geometry, so we must now ask: how do we extend this logic to the integration of functions?
You will recall that much earlier in the semester, when we first defined the Riemann integral of a positive function f over a domain A, we defined it strictly in terms of the measure of its hypograph.
What is the hypograph geometrically? It is the $(n+1)$-dimensional solid region situated directly beneath the graph of the function and directly above the base domain A. If we wish to integrate a function $f(x,y)$ over a product space $\mathbb{R}^k \times \mathbb{R}^{n-k}$, we are effectively seeking the $(n+1)$-dimensional volume of this overarching solid block.
Therefore, integrating a function is fundamentally identical to measuring a physical shape. We can simply apply Cavalieri's Principle directly to this $(n+1)$-dimensional hypograph. By taking vertical slices of this solid, we reduce a massive multi-dimensional volume calculation into an iterated sequence of much simpler lower-dimensional integrals. This elegant corollary is what we call Fubini's Theorem.

Board State & Physical Actions: Board State 10: Recalling the Hypograph Connection

The professor moves to the freshly cleaned center board. He writes down the formal definition of an integral via the hypograph, establishing the crucial bridge between the pure geometry of Cavalieri and the operational calculus of Fubini.

Integral as a Measure of the Hypograph

Let $f : A \to \mathbb{R}_{\geq 0}$ be a Riemann integrable function over a Jordan measurable set $A \subset \mathbb{R}^n$. The hypograph of f is defined as the $(n+1)$-dimensional set:

$$\mathrm{Hypo}(f) = \{(x,z) \in \mathbb{R}^n \times \mathbb{R} \mid x \in A,\ 0 \leq z \leq f(x)\}$$

By our foundational definition, the integral of the function is identically the measure of this set:

$$\int_A f(x)\,dx = \mu_{n+1}(\mathrm{Hypo}(f))$$

Insight: The Geometric Unification

This is the grand unification of the lecture. By framing the integral not as an abstract analytic operation, but as a literal physical volume (the hypograph), the professor allows the students to seamlessly apply the "potato slicing" intuition they just mastered with Cavalieri's Principle directly to abstract function integration.

20 Statement of Fubini's Theorem

Cleaned Spoken Protocol [Time: 34:15 - 38:00]

Let us formalize this into a proper theorem. Suppose we partition our base domain space $\mathbb{R}^n$ into a Cartesian product $\mathbb{R}^k \times \mathbb{R}^{n-k}$, utilizing coordinate vectors x and y respectively. Let A be our domain of integration, and $f(x,y)$ be our continuous, integrable function hovering above it.
If we freeze a base coordinate $x \in \mathbb{R}^k$, treating it strictly as a constant, we can restrict our function to the corresponding vertical slice of the domain, A_x. We shall denote this restricted, lower-dimensional function as $f_x(y)$.
Under Fubini's Theorem, assuming the requisite integrability conditions hold over the individual slices, the total n-dimensional integral of f over the entire domain A is exactly equivalent to the iterated integral!
Operationally, what does this mean? You first integrate the restricted function $f_x(y)$ with respect to y over the slice A_x. Because x was held constant, the result of this inner integral is a new marginal function purely in terms of x. Subsequently, you integrate this resulting marginal function with respect to x over the base projection of the domain.

Board State & Physical Actions: Board State 11: Formalizing the Sliced Function

The professor meticulously writes out the notation for the restricted function $f_x(y)$. He taps the board with his chalk under the x variable to emphasize to the students that x must be treated purely as a constant scalar during the inner integration step. He then writes the full statement of Fubini's Theorem in bright orange chalk.

Function Restriction and Iteration

Given a function $f : A \to \mathbb{R}$ where the domain $A \subset \mathbb{R}^k \times \mathbb{R}^{n-k}$.
For a fixed, constant $x \in \mathbb{R}^k$, define the slice domain $A_x = \{y \mid (x,y) \in A\}$.
We define the restricted slice function $f_x : A_x \to \mathbb{R}$ as:

$$f_x(y) = f(x,y) \quad \text{(treating } x \text{ strictly as a constant)}$$

Fubini's Theorem

Let $A = X \times Y$ be a compact, Jordan measurable domain, and $f : A \to \mathbb{R}$ be integrable. The multiple integral can be evaluated sequentially as an iterated integral:

$$\int_A f(x,y)\, d(x,y) = \int_X \left(\int_{A_x} f(x,y)\, dy \right) dx = \int_Y \left(\int_{A_y} f(x,y)\, dx \right) dy$$

21 Visualizing the Iterated Integral

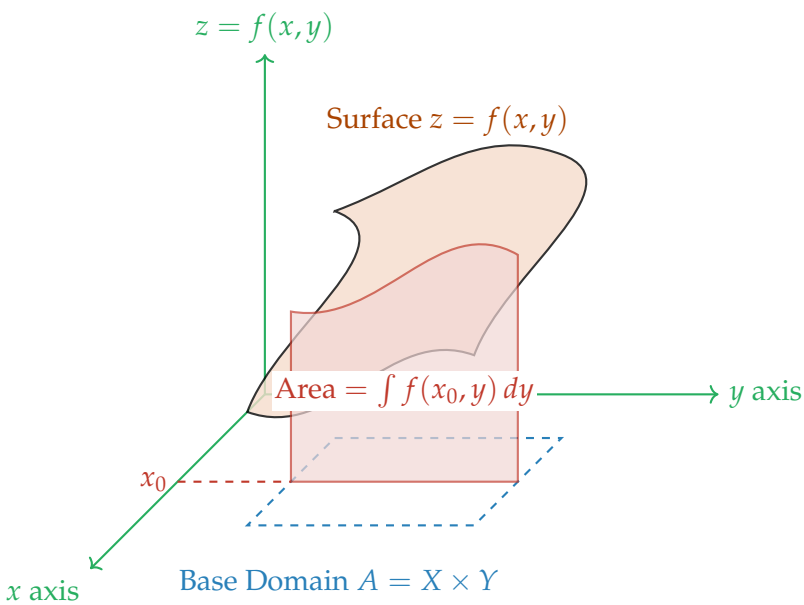
Cleaned Spoken Protocol [Time: 38:00 - 42:30]

To truly grasp the mechanics of Fubini's Theorem and prevent algebraic mistakes, one must visualize the sequential nature of this volume accumulation.
When we evaluate the inner integral—integrating $f(x, y)$ strictly with respect to y while holding x frozen at some value x_0—what are we calculating geometrically? We are calculating the exact 2D cross-sectional area of the solid hypograph at that specific coordinate x_0.
This inner integral effectively collapses the massive $(n + 1)$-dimensional solid into a one-dimensional density function residing on the base axis. The outer integral then sweeps this cross-sectional area across the entire length of the domain X, accumulating those differential slices to arrive at the total total measure of the solid.
The symmetry of the theorem ensures that the order of accumulation—whether sweeping left-to-right along the x-axis or front-to-back along the y-axis—is irrelevant to the final numerical volume, provided your integration boundaries A_x and A_y are properly accounted for.

Board State & Physical Actions: Board State 12: The Hypograph Cross-Section

The professor moves to the right board to draw a masterful 3D perspective of a curved surface floating above a rectangular domain in the xy-plane. He uses red chalk to shade a specific 2D planar cross-section beneath the surface, labeling its area as the exact result of the inner integral, $\int f(x_0, y)dy$.

Geometric Visualization of Fubini's Theorem



Why the Order Matters for Limits

The visual clarifies a crucial stumbling block for students: why the bounds of integration change when you swap dx and dy. The red slice represents the inner integral: evaluating the area of the 2D plane at a constant x_0. As x_0 moves along the x-axis, the boundaries of that slice might grow or shrink if the base domain A is not a perfect rectangle (e.g.,

a triangle or a circle). The outer integral then integrates this continuously changing 1D area function along the x-axis to get the final 3D volume.

Cleaned Spoken Protocol [Time: 42:30 - 45:00]

So, Fubini's Theorem is not magic; it is simply Cavalieri's Principle applied to the volume under a graph. By treating one variable as a constant, you reduce a multivariable problem into a sequence of single-variable calculus problems that you already know how to solve from Analysis I.
We will spend the next lecture doing heavy computational examples of this, swapping the bounds of integration, and looking at how this interacts with our Change of Variables theorem from last week. Please review your single-variable integration techniques tonight, because you will be doing them twice as much now. Have a good afternoon, everyone.

End of Lecture

The professor sets down his chalk and addresses the class. The students begin packing up. The lecture concludes exactly at 45:00, having successfully bridged the geometric theory of measures to the operational computation of multiple integrals.